A Report

on

Global Temperature Trend Analysis using Machine Learning

carried out as part of the course Machine Learning Lab

Submitted by

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Abstract

This project presents a comprehensive analysis of global temperature anomalies using the Berkeley Earth Land–Ocean dataset, which contains monthly temperature anomaly records along with associated uncertainty values from 1850 onwards. The objective of the study is to analyze temporal temperature patterns and develop predictive and analytical models to better understand variations in global temperature over time.

The dataset, provided in a structured text format, is first loaded and processed into a tabular form with clearly defined attributes such as land, ocean, and combined temperature anomalies. Missing values present in the dataset are handled using column-wise mean imputation to ensure continuity in analysis. Exploratory Data Analysis (EDA) is then performed using statistical summaries, histograms, distribution plots, pairwise relationships, and correlation heatmaps to understand the distribution and relationships between different variables.

To enhance the dataset for modeling, feature engineering techniques are applied. A datetime feature is constructed from year and month, followed by the creation of a time index. Additionally, lag features (previous month and previous year) and rolling statistical features (12-month rolling mean and standard deviation) are introduced to capture temporal dependencies and trends in the data.

The study employs multiple machine learning models, including Support Vector Regressor (SVR), Decision Tree Regressor, and Random Forest Regressor, to predict the combined global temperature anomaly. A time-based train-test split is used to preserve the temporal nature of the dataset. Model performance is evaluated using metrics such as Mean Absolute Error (MAE) and R^2 score, with Random Forest demonstrating strong predictive capability.

Further analysis includes residual-based anomaly detection, where extreme deviations between actual and predicted values are identified using a statistical threshold. These anomalies are evaluated using ROC curve analysis and confusion matrix visualization. In addition, unsupervised clustering techniques such as K-Means, Hierarchical Clustering, and DBSCAN are applied to identify inherent patterns and groupings within the temperature data.

Overall, this project demonstrates the effective use of data preprocessing, feature engineering, machine learning modeling, and clustering techniques to analyze and interpret global temperature anomaly data, providing meaningful insights into temperature variation patterns over time.

1. Introduction

1.1 Background

Over the past century, global temperatures have shown a consistent upward trend, largely driven by increased greenhouse gas emissions resulting from human activities such as industrialization, deforestation, and the extensive use of fossil fuels. This rise in temperature is a fundamental indicator of climate change and has significant consequences on natural systems, including melting glaciers, rising sea levels, and changing weather patterns. Analyzing historical temperature data is essential to understand these long-term trends and variations, and with the advancement of data analysis and machine learning techniques, it has become possible to extract meaningful insights and identify patterns more efficiently, aiding in the study and monitoring of global warming.

1.2 Problem Statement

The project focuses on analyzing global temperature anomaly data over an extended historical period to identify patterns, trends, and variations in temperature. The dataset contains monthly temperature anomalies along with associated uncertainty values, as well as aggregated statistics such as annual and multi-year averages. Due to the temporal nature and scale of the dataset, direct interpretation is not straightforward.

The key problem is to process and analyze this data in a structured manner so that meaningful insights can be extracted. This includes organizing the data, handling missing values, and applying analytical and machine learning techniques to study how temperature values change over time. Additionally, the project aims to model these variations, detect unusual deviations, and identify underlying patterns within the data.

1.3 Importance of The Topic

Analyzing global temperature data is important for understanding long-term changes in climate behavior. Temperature anomalies provide a consistent way to measure deviations relative to a baseline period, making it possible to compare values across different time periods. With datasets spanning over a century, it becomes essential to use systematic and computational approaches to analyze such large volumes of data effectively.

This project is significant because it demonstrates how data analysis and machine learning techniques can be applied to real-world climate datasets. Through exploratory data analysis, important characteristics such as distribution, relationships, and variability in temperature data can be observed. Feature engineering further enhances the dataset by capturing temporal dependencies, enabling more effective modeling.

The use of machine learning models allows for prediction and better understanding of temperature trends, while anomaly detection techniques help identify unusual or extreme deviations in the data. Additionally, clustering methods provide a way to group similar patterns, offering deeper insight into the structure of temperature variations.

Overall, the project highlights the role of data-driven approaches in analyzing complex environmental data and extracting meaningful insights from long-term temperature records.

2. Dataset & Exploratory Data Analysis

2.1 Dataset Overview

The dataset used in this project is the **Berkeley Earth Land–Ocean Temperature dataset**, which contains historical records of global temperature anomalies. The data is provided in a text file format and includes monthly temperature observations starting from the year 1850.

Each record in the dataset represents temperature anomaly values rather than absolute temperatures. These anomalies are calculated relative to a baseline period (January 1951 to December 1980), allowing consistent comparison of temperature variations over time. Along with anomaly values, the dataset also includes corresponding uncertainty estimates, providing a measure of confidence in the recorded data.

The dataset contains multiple attributes, including:

- Year and Month (temporal features)
- Land temperature anomaly
- Ocean temperature anomaly
- Combined land–ocean temperature anomaly
- Uncertainty values for each measurement

In addition to monthly data, the dataset also provides aggregated values such as annual averages and multi-year moving averages (5-year, 10-year, and 20-year), which help in analyzing long-term trends and reducing short-term fluctuations.

Overall, the dataset is well-structured and suitable for time-series analysis, enabling the study of temperature variations, trend identification, and application of machine learning techniques.

2.2 Target Variable Distribution

The target variable in this project is the **combined land–ocean temperature anomaly**, which represents the deviation of global temperature relative to a fixed baseline period. This variable is analyzed to understand its distribution and suitability for predictive modeling.

The distribution is examined using histogram and density (KDE) plots, which provide insight into the spread and concentration of values. The target variable exhibits a continuous distribution with values spanning both negative and positive ranges. Earlier observations in the dataset are primarily concentrated in the negative range, while later observations show a shift toward positive values.

The distribution indicates that most values lie within a moderate range, with fewer extreme values at the tails. The shape of the distribution reflects variability in temperature anomalies over time and suggests the presence of an overall shift in values rather than a uniform spread.

Analyzing the distribution of numerical features is analyzed, with particular focus on the combined temperature anomaly for understanding its behavior, identifying the range and concentration of values, and ensuring that it is appropriate for regression-based machine learning models.

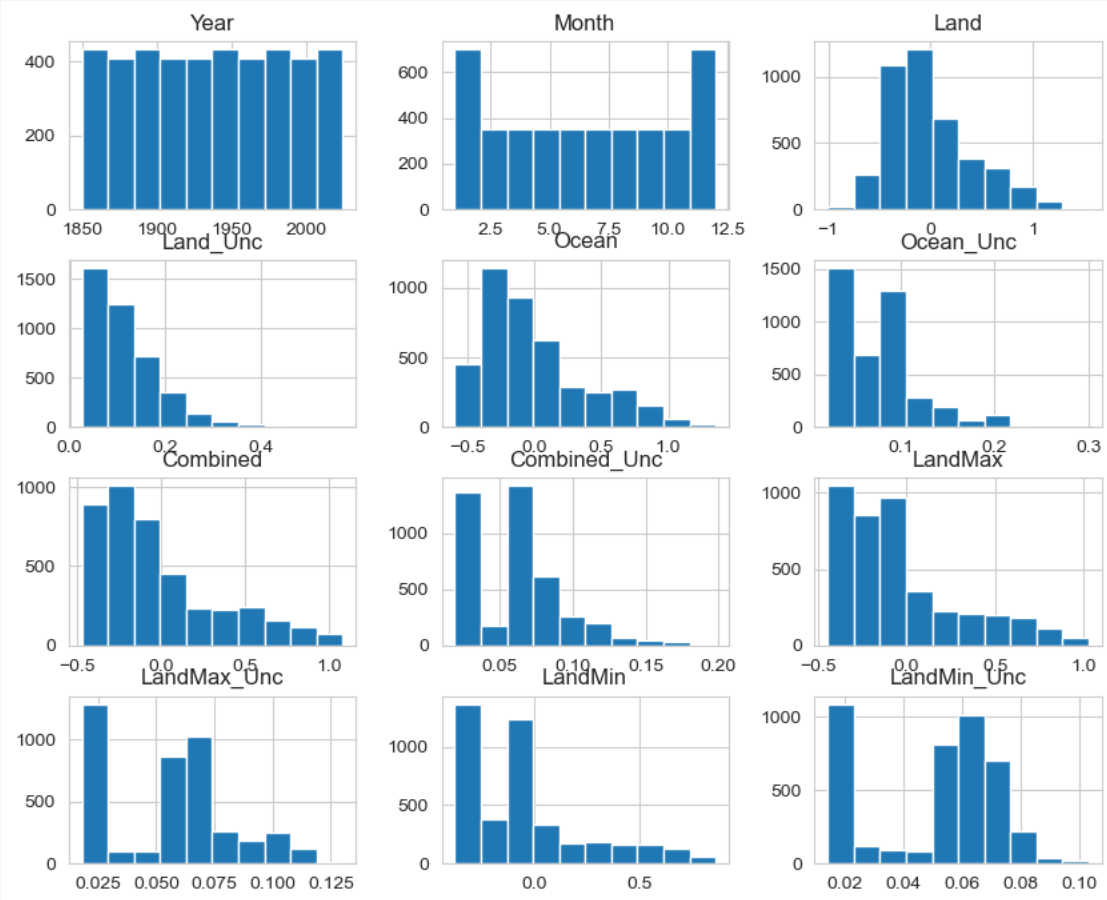


Figure 1: Distribution of Numerical Features in the Global Temperature Dataset

2.3 Feature Correlation Analysis

The correlation heatmap illustrates the relationships between different temperature variables and engineered features in the dataset. Strong positive correlations can be observed between related temperature variables such as land, ocean, and combined temperature anomalies, indicating that these features vary consistently with each other.

The combined temperature anomaly, which serves as the target variable, shows high correlation with features such as lag variables (lag_1, lag_12) and rolling mean (rolling_mean_12). This indicates that past temperature values and aggregated trends play a significant role in predicting current temperature values.

Time-based features such as the time index also exhibit a noticeable correlation with temperature variables, reflecting the temporal progression present in the dataset. On the other hand, uncertainty-related features generally show weaker or negative correlations with temperature variables, suggesting that they contribute differently to the dataset.

Overall, the heatmap highlights important relationships and validates the relevance of engineered features used for modeling, particularly lag and rolling features, which capture temporal dependencies in the data.

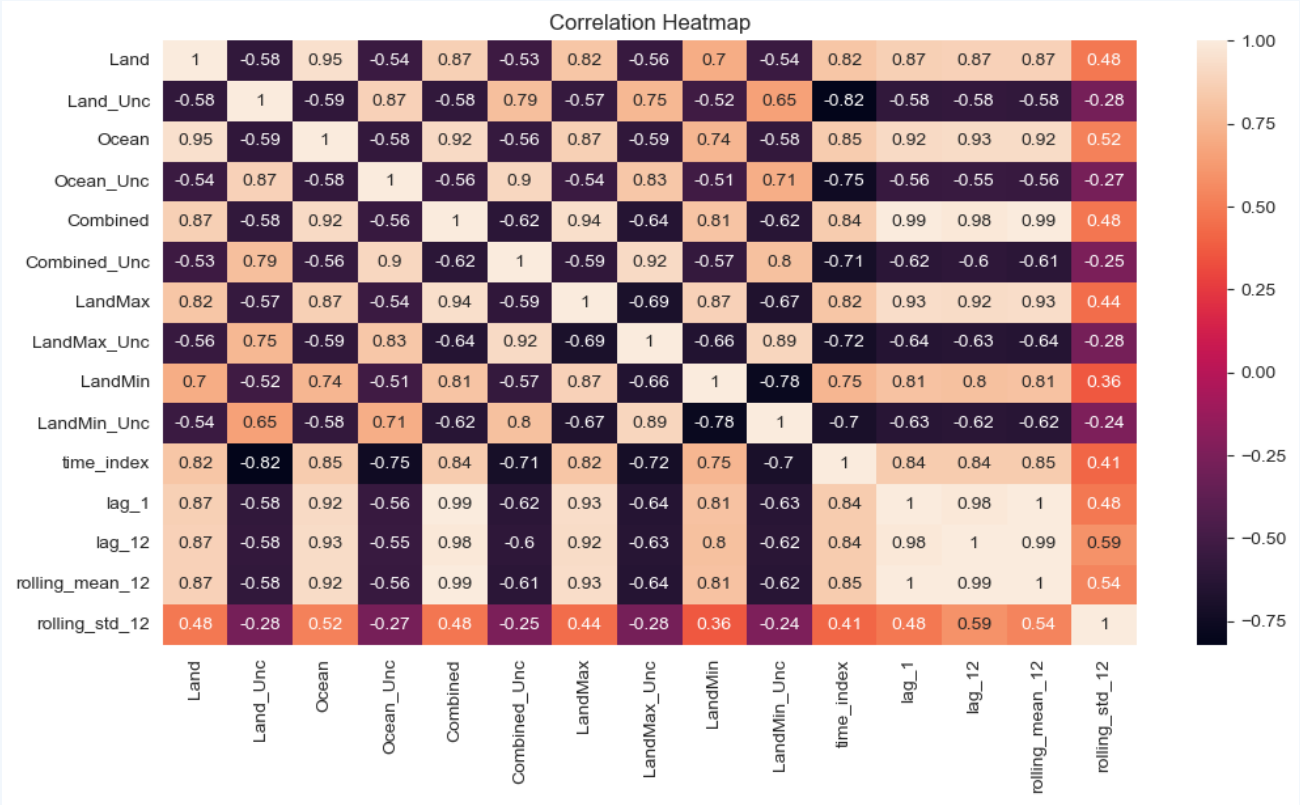


Figure 2: Pearson Correlation Heatmap

In addition, the heatmap assists in identifying the most relevant features for modeling by highlighting strong relationships with the target variable. Features with higher correlation values are likely to contribute more effectively during prediction, while weaker correlations indicate lesser influence. This supports the feature engineering process and ensures that meaningful variables are used in building accurate models.

3. Methodology

3.1 Overview

The methodology of this project follows a structured pipeline to analyze global temperature anomaly data and build predictive models. The process includes data loading, preprocessing, exploratory analysis, feature engineering, model development, and evaluation. Each step is designed to transform raw data into meaningful insights and accurate predictions..

3.1.1 Data Loading

The dataset is loaded from a text file containing historical global temperature anomaly records. It includes features such as year, month, land temperature anomaly, ocean temperature anomaly, combined temperature anomaly, and uncertainty values. The data is converted into a structured tabular format for further analysis.

3.1.2 Data Processing

Before analysis, the dataset is prepared by handling missing values and ensuring consistency. Missing values present in the dataset are replaced using mean imputation to maintain continuity and avoid data loss. This step ensures that the dataset is suitable for further processing and modeling.

3.1.3 Exploratory Data Analysis (EDA)

Exploratory analysis is performed to understand the distribution and relationships within the dataset. This includes generating histograms to study feature distributions and correlation heatmaps to analyze relationships between variables. These visualizations help in identifying patterns and dependencies in the data.

3.1.4 Feature Engineering

Exploratory analysis is performed to understand the distribution and relationships within the dataset. This includes generating histograms to study feature distributions and correlation heatmaps to analyze relationships between variables. These visualizations help in identifying patterns and dependencies in the data.

3.1.5 Model Development

Multiple machine learning models are applied to predict the combined temperature anomaly. These include Support Vector Regressor (SVR), Decision Tree Regressor, and Random Forest Regressor. The dataset was split into training and testing sets for model evaluation..

3.1.6 Model Evaluation

The performance of the models is evaluated using metrics such as Mean Absolute Error (MAE) and R^2 score. These metrics help in assessing the accuracy and effectiveness of each model in predicting temperature anomalies.

3.1.8 Clustering

Unsupervised learning techniques such as K-Means, Hierarchical Clustering, and DBSCAN are applied to group similar patterns in the data. This helps in identifying underlying structures and patterns in temperature variations.

4. Results & Evaluation

4.1 Model Performance Summary

Three machine learning models—Support Vector Regressor (SVR), Decision Tree Regressor, and Random Forest Regressor—were evaluated using Mean Absolute Error (MAE):

Model	MAE
SVR	0.132325
Decision Tree	0.245477
Random Forest	0.267033

Table 2: Comparative performance of all three classifiers

The SVR model achieves the lowest MAE, indicating the best prediction accuracy among the models. Decision Tree and Random Forest exhibit higher error values, suggesting comparatively weaker performance on this dataset.

4.2 Prediction Analysis

The actual vs predicted plot for the Random Forest model shows that while the model attempts to follow the overall trend, it fails to capture the increasing pattern of temperature anomalies effectively. The predicted values remain relatively flat compared to the actual values, especially in later regions of the dataset.

This indicates that the Random Forest model is not effectively modeling the temporal progression present in the data, leading to higher prediction errors.

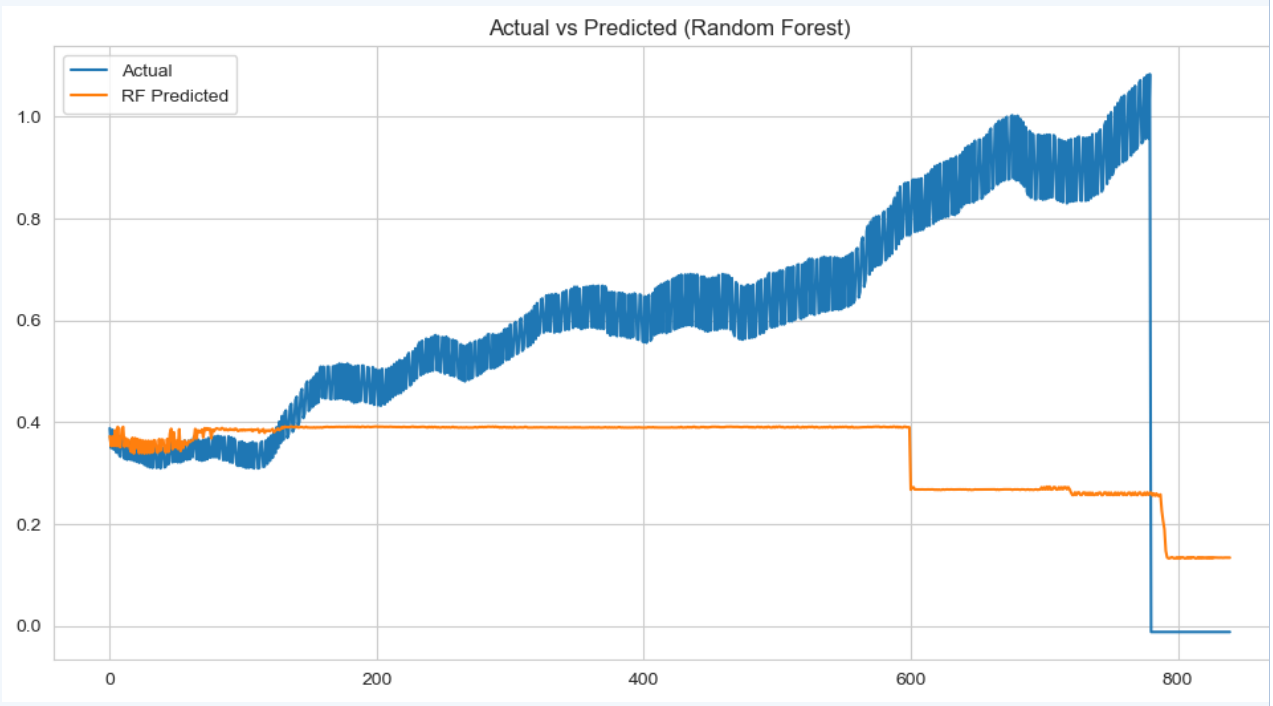


Figure 3: Actual vs Predicted Plot of Random Forest Model

4.3 Residual Analysis

The residual plot shows the distribution of prediction errors across the dataset. The presence of large and spread-out residual values indicates that the model errors are not uniformly distributed and increase in certain regions.

This suggests that the model struggles to generalize across different segments of the data, particularly where the trend changes.

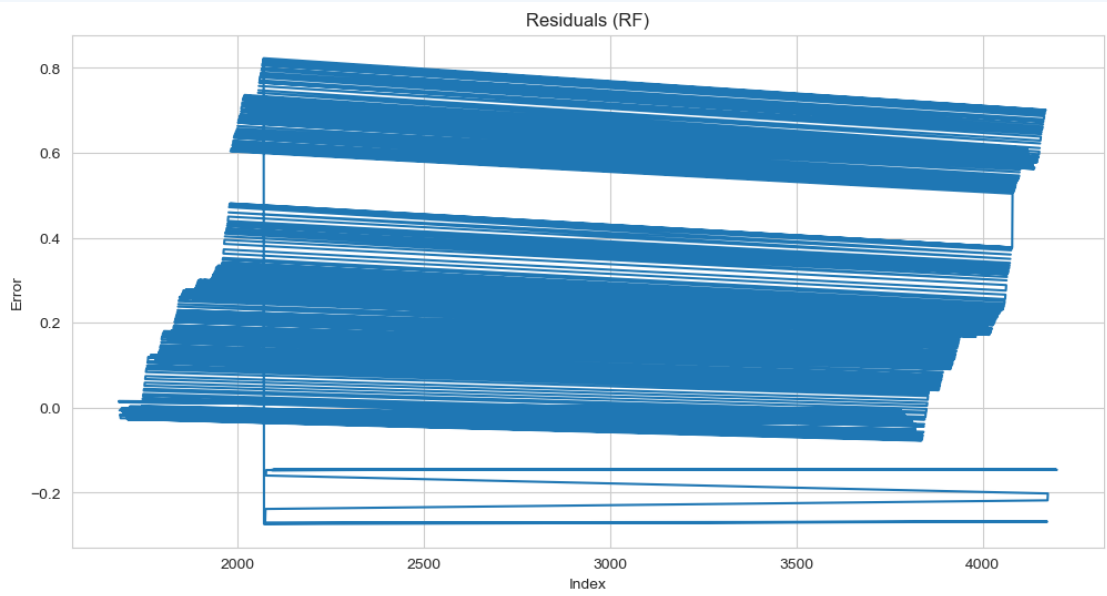


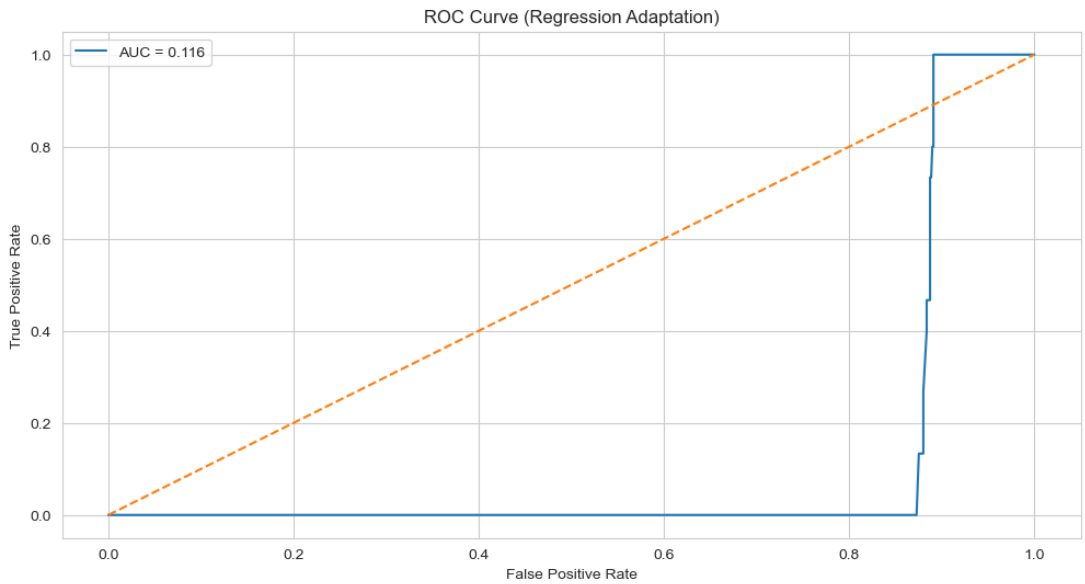
Figure 8: Residual Plot showcasing distribution of prediction errors across dataset.

4.4 Anomaly Detection and ROC Analysis

Anomaly detection was performed using residual-based classification. The ROC curve was plotted to evaluate performance, and the Area Under Curve (AUC) was calculated as:

AUC = 0.1156

This very low AUC value indicates that the model performs poorly in distinguishing between normal and anomalous observations. The ROC curve lies close to or below the random baseline, suggesting weak classification capability.



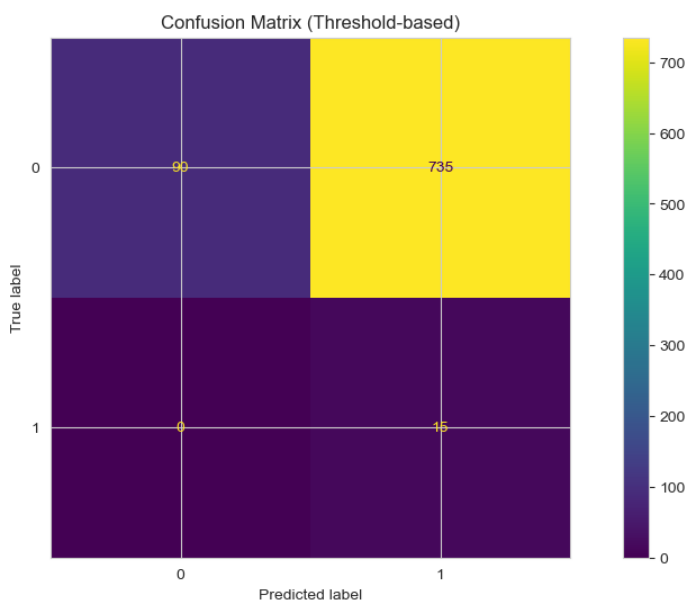
4.5 Confusion Matrix Analysis

The confusion matrix shows the classification results:

- True Negatives (TN): 90
- False Positives (FP): 735
- False Negatives (FN): 0
- True Positives (TP): 15

This indicates:

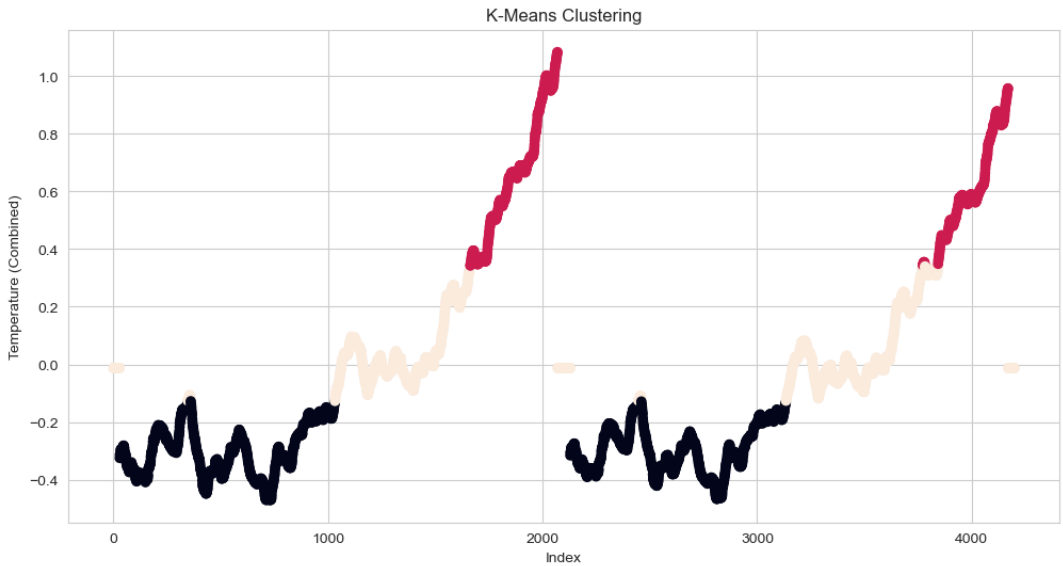
- A very high number of false positives.
- The model tends to incorrectly classify many normal points as anomalies
- Although actual anomalies are detected (TP), overall classification performance is poor.



4.6 Clustering Results

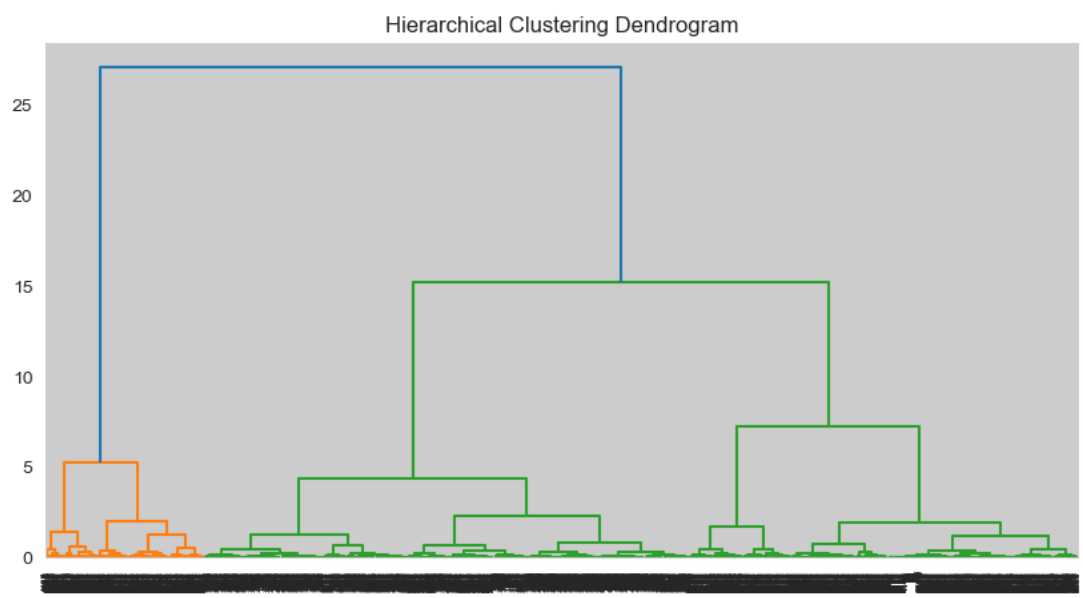
(a) K-Means Clustering

K-Means clustering groups the data into distinct clusters based on temperature values. The clusters show separation between lower and higher temperature ranges, reflecting different phases of temperature variation. For k=3, Graph is shown below:-



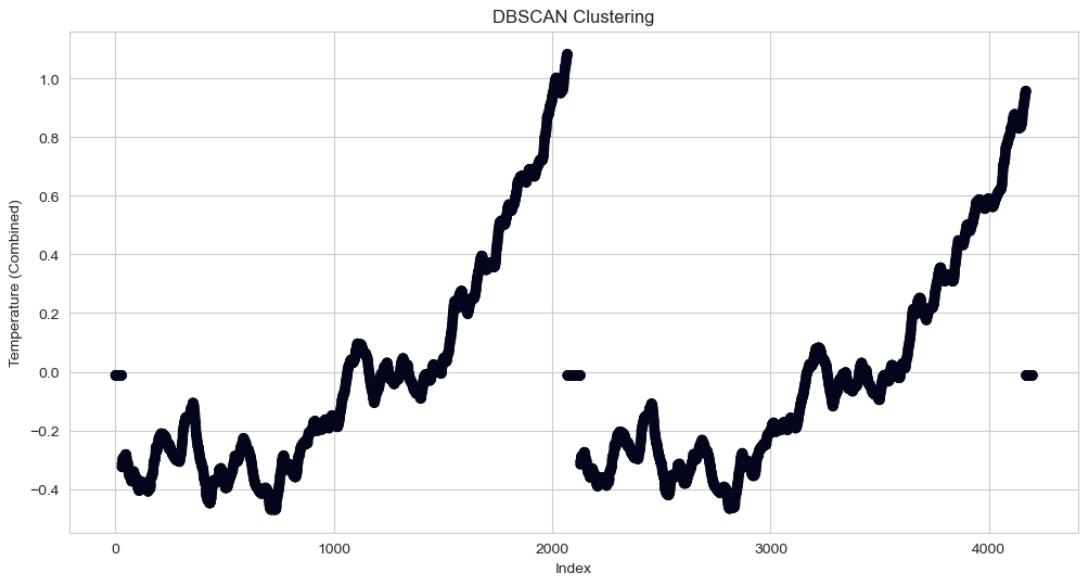
(b) Hierarchical Clustering

The dendrogram illustrates how data points are grouped at different levels of similarity. It shows hierarchical relationships and merging patterns between clusters.



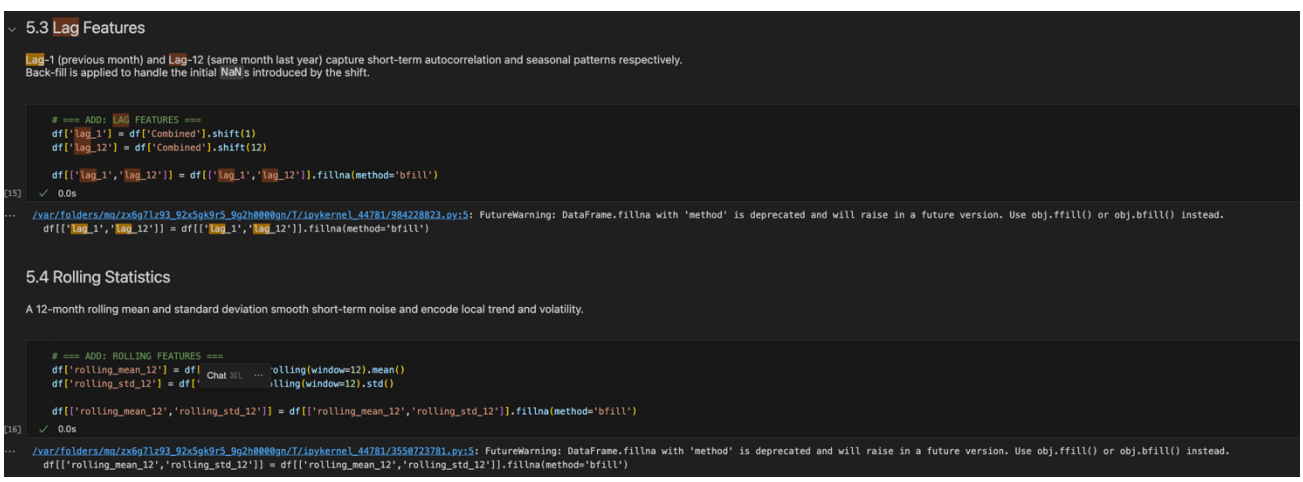
(c) DBSCAN Clustering

DBSCAN identifies dense regions in the data and highlights noise points. In this case, the clustering shows limited separation, indicating that density-based clustering does not strongly distinguish patterns in the dataset.



4.7 Feature Effectiveness

Feature engineering plays a critical role in the analysis. Lag features and rolling statistics capture temporal dependencies, which are essential for modeling time-based data such as temperature anomalies. However, model performance indicates that not all models are equally capable of utilizing these features effectively.



5.3 Lag Features

Lag-1 (previous month) and Lag-12 (same month last year) capture short-term autocorrelation and seasonal patterns respectively.
Back-fill is applied to handle the initial NaNs introduced by the shift.

```
# === ADD: LAG FEATURES ===
df['lag_1'] = df['Combined'].shift(1)
df['lag_12'] = df['Combined'].shift(12)

df[['lag_1','lag_12']] = df[['lag_1','lag_12']].fillna(method='bfill')
```

[15] ✓ 0.0s

```
/var/folders/mq/zx6g7lz93_92x5gk9r5_9g2h0000gn/T/ipykernel_44781/984228823.py:5: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.
  df[['lag_1','lag_12']] = df[['lag_1','lag_12']].fillna(method='bfill')
```

5.4 Rolling Statistics

A 12-month rolling mean and standard deviation smooth short-term noise and encode local trend and volatility.

```
# === ADD: ROLLING FEATURES ===
df['rolling_mean_12'] = df[ Chat ⌘L ··· olling(window=12).mean()
df['rolling_std_12'] = df[' illing(window=12).std()

df[['rolling_mean_12','rolling_std_12']] = df[['rolling_mean_12','rolling_std_12']].fillna(method='bfill')
```

[16] ✓ 0.0s

```
/var/folders/mq/zx6g7lz93_92x5gk9r5_9g2h0000gn/T/ipykernel_44781/3550723781.py:5: FutureWarning: DataFrame.fillna with 'method' is deprecated and will raise in a future version. Use obj.ffill() or obj.bfill() instead.
  df[['rolling_mean_12','rolling_std_12']] = df[['rolling_mean_12','rolling_std_12']].fillna(method='bfill')
```

4.8 Overall Evaluation

The overall evaluation highlights clear differences in model performance and effectiveness across different analytical components. Among the models used, the Support Vector Regressor achieved the lowest Mean Absolute Error, indicating superior predictive accuracy compared to Decision Tree and Random Forest models. This suggests that SVR is better suited for capturing the underlying patterns in the dataset.

However, analysis of the Random Forest predictions and residuals shows that certain models struggle to capture the temporal progression and variability present in temperature anomaly data. The presence of large residuals and deviations indicates limitations in modeling complex time-dependent trends.

The anomaly detection results further reflect these challenges. The low AUC value (0.1156) and the high number of false positives in the confusion matrix indicate that the model is not effective in distinguishing between normal and anomalous observations. This suggests that the chosen approach for anomaly detection may not be well-suited for this dataset.

Clustering techniques provided additional insights into the structure of the data, revealing grouping patterns based on temperature values. However, the lack of strong separation in some methods indicates that temperature variations may not form clearly distinct clusters.

Overall, while the project successfully applies multiple machine learning and analytical techniques, the results demonstrate that modeling time-dependent climate data is complex. The effectiveness of different approaches varies significantly, highlighting the importance of model selection, feature design, and appropriate evaluation methods.

5. Conclusion

This project presented a comprehensive analysis of global temperature anomaly data using data analysis and machine learning techniques. The dataset was systematically processed and explored to understand underlying patterns, followed by feature engineering to capture temporal dependencies through lag variables and rolling statistics. These steps enabled the application of predictive models on structured time-series data.

Among the models evaluated, the Support Vector Regressor achieved the lowest Mean Absolute Error, indicating better predictive performance compared to Decision Tree and Random Forest models. The prediction and residual analysis highlighted that certain models struggled to effectively capture the increasing trend and variability in temperature anomalies.

The anomaly detection results, supported by the ROC curve and confusion matrix, showed limited effectiveness, as reflected by the low AUC value and high number of false positives. Clustering techniques provided additional insights into the grouping and structure of the data, although clear separations were not always observed.

Overall, the project demonstrates that while machine learning techniques can be applied to analyze climate data and extract meaningful patterns, modeling time-dependent data such as temperature anomalies remains challenging and requires careful feature design and model selection.

7. Future Scope

- Apply advanced time-series models such as LSTM or ARIMA to better capture temporal dependencies.
- Improve anomaly detection using more robust techniques or optimized threshold selection.
- Perform hyperparameter tuning to enhance model performance and reduce prediction error.
- Incorporate additional climate variables (e.g., CO_2 levels, humidity) for more comprehensive analysis.
- Explore feature selection methods to identify the most impactful variables for prediction.

8. References

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